UM HACKATHON 2026 · TEAM SHAKEHAND

Z.AI Integrated Flood Emergency Portal

An Agentic AI-Driven Ecosystem for Real-Time Rescue Coordination and Survival Navigation.

Case Study: Sri Muda 2021

In Dec 2021, during the catastrophic Taman Sri Muda floods, 120,000+ victims were stranded. The failure was not a lack of resources, but a **communication blackout**.

Rescue agencies were buried under thousands of disjointed WhatsApp and Twitter messages, leading to a 48-hour delay in critical life-saving missions.

Z.AI bridges the gap between flood chaos and tactical rescue intelligence.

The Problem Statement

Dynamic Inundation

Flash floods change street topography in minutes. Static maps become death traps for responders.

Human Bottleneck

Dispatchers can only process one call at a time. High-volume emergencies lead to dispatcher fatigue and error.

Asset Gridlock

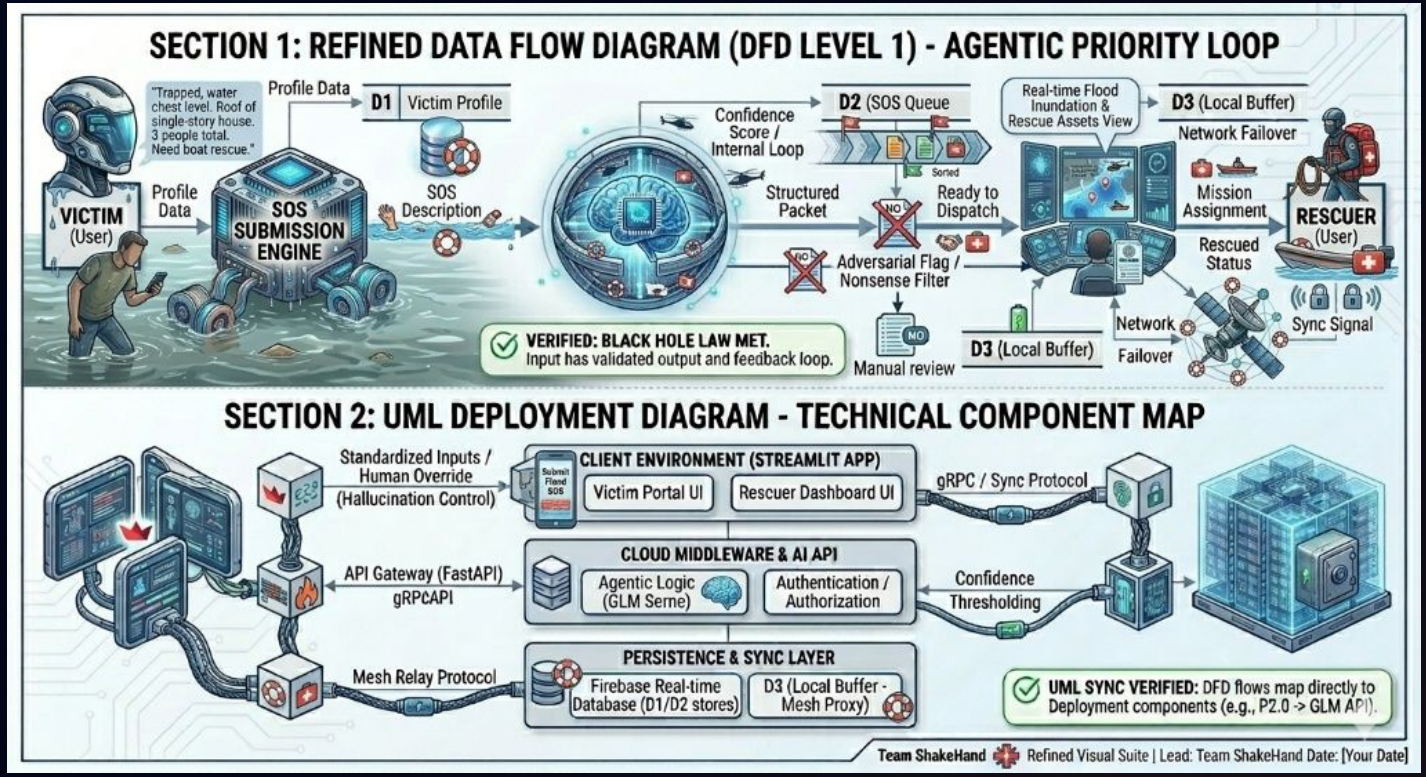
Without priority sorting, rescue boats are dispatched to knee-deep water while chest-deep victims wait.

System Under Test (SUT)

Z.AI Core Architecture

The Z.AI Integrated Portal consists of two interconnected modules powered by a central **Agentic Priority Engine (P2.0)**.

- **Victim App:** Accessible SOS broadcasting.
- **Rescue Dashboard:** Command & Control.
- **Real-time Backend:** Firebase NoSQL Sync.
- **AI Layer:** Hallucination-resistant LLM agents.



Module 1: Victim Portal

User-Centric SOS Interface

Designed for high-stress scenarios where every second and every kilobyte of data counts.

- **Standardized Inputs:** Radio buttons for Water Level (Knee, Hip, Chest) and Group Size.
- **Escape Routing:** AI calculates paths to high ground.
- **Mesh Buffering:** Works in 2G/Mesh environments.
- **Visual Beacon(Dead Phone Protocol) :** Flashes screen to assist final-mile rescue.

Module 2: Rescue Dashboard

The Commander's Lens

A high-fidelity monitoring station for emergency services to manage assets and missions.

- **Hotzone Visualization:** Heatmaps derived from SOS density.
- **Mission Lifecycle:** Dispatched → On-Site → Rescued.
- **Asset Sync:** Real-time Firebase handshake with field units.

Agentic Priority Engine P2.0

P0: CRITICAL
Chest-level water, medical trauma, or roof-stranded victims. **Immediate response.**

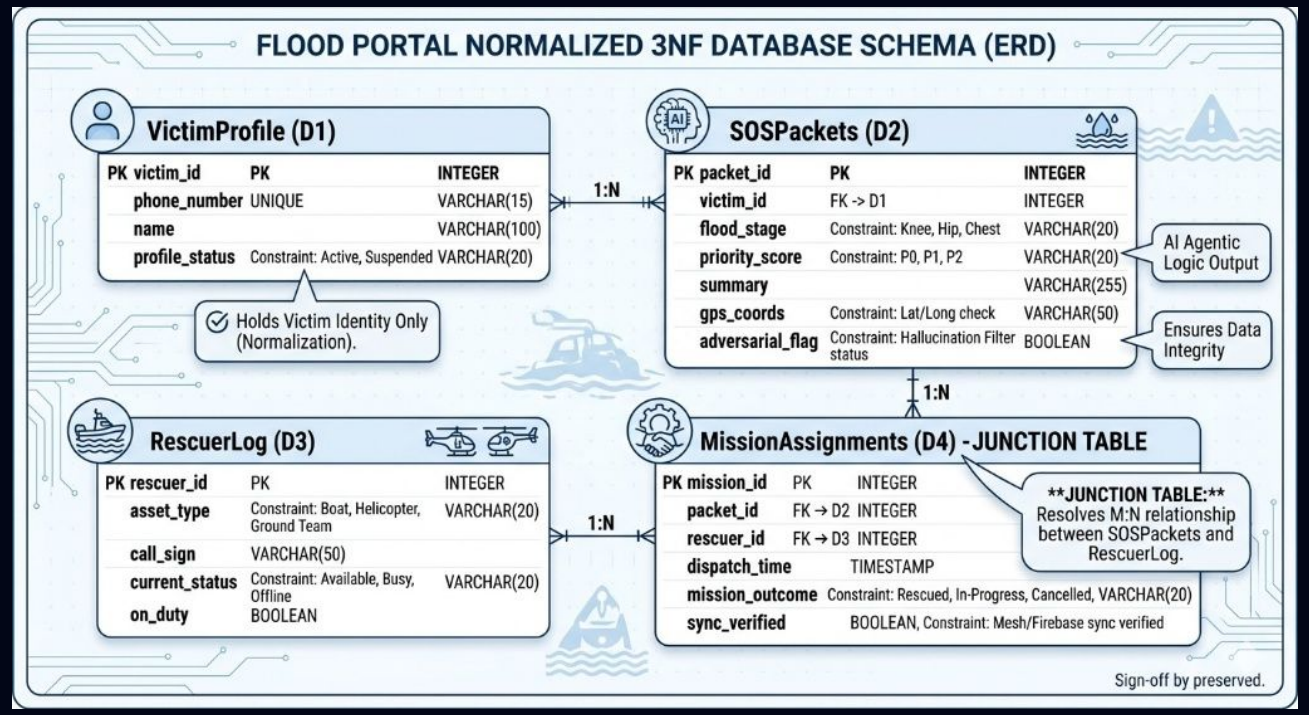
P1: URGENT
Hip-level water, supplies running low, elderly occupants. **Priority response.**

P2: MONITORING
Knee-level water, relocation requests, property damage. **Scheduled response.**

Powered by multi-agent prompt engineering to filter adversarial "noise" and hallucinated distress signals.

Database Analysis: 3NF Normalization

Table	Description
D1: VictimProfile	PII storage (Name, Phone). PK: victim_id
D2: SOSPackets	The signal data. FK: victim_id
D3: RescuerLog	Directory of assets (Boat, Heli).
D4: MissionLog	Junction Table resolving M:N Signal-to-Rescuer.



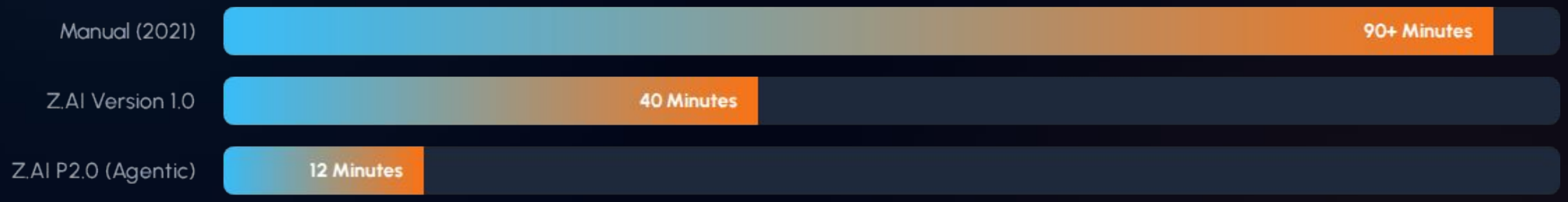
AI Safety: Hallucination Mitigation

- **Geospatial Grounding Audit:** Cross-references AI-parsed locations with GIS data to block signals mapping to non-existent addresses.
- **Response Validation Layer (Logic Gate):** Flags logical contradictions (e.g., Marking someone “safe” when their water level is “Chest-High”)
- **80% Confidence Threshold:** Any AI summary falling below 80% certainty is routed to a human dispatcher for manual verification.
- **Negative Prompt Engineering:** Strictly forbids the AI from "inventing" details not present in the raw victim SOS input.

Validation: Prompt/Response Test Pairs

Test ID	Prompt Input (Victim)	Expected AI Output	Status
AI-01	4 victims trapped, water above knees, severe bleeding & hypothermia.	Priority: P0 | Dispatch: Swift Water Rescue, Paramedics.	PASSED
AI-02	1 victim safe on 2nd floor, water above hips, requesting food.	Priority: P2 | Dispatch: Standard Evac, Supply Boat.	PASSED
AI-03	"xdfg sdfsdf water is blue" (Simulated Panic/Nonsensical Input)	Priority: P-Review | Adversarial Flag: TRUE (Routed to Manual Review).	PASSED

Efficiency Metrics: Speed to Rescue



Simulated response time from SOS receipt to Rescuer Dispatch using the Agentic Priority Engine.

SYSTEM EVOLUTION ROADMAP

A strategic timeline for expanding Z.AI from a rapid-response pilot to a resilient global disaster framework.

RESILIENCE

Phase 1

Satellite Mesh integration for total communication blackout resilience.

INTERVENTION

Phase 2

Drone-based first aid delivery and critical supply drops to isolated victims.

PREDICTION

Phase 3

Predictive AI flood inundation modeling for pre-emptive relocations.

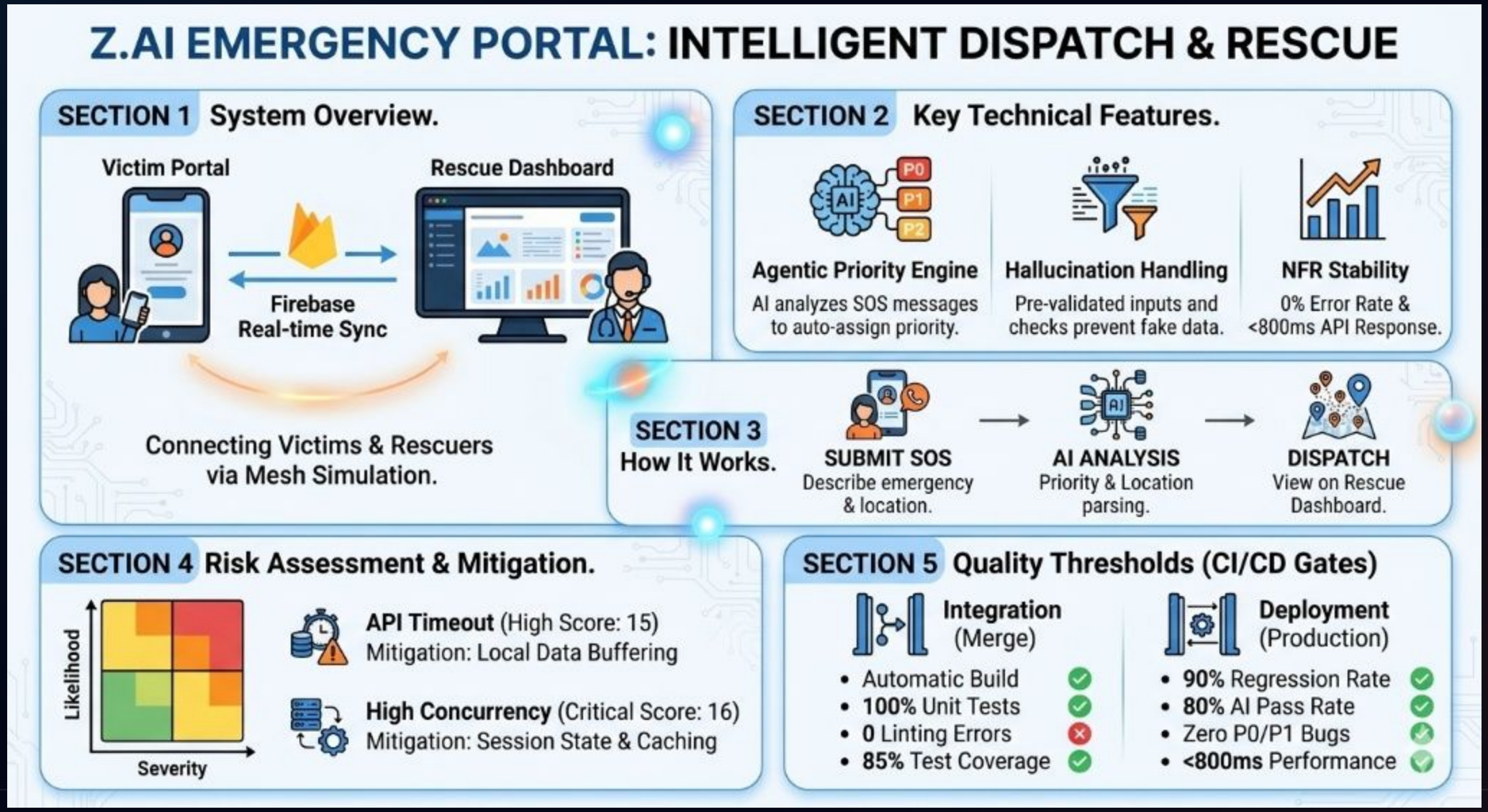
EXPANSION

Phase 4

Multi-language NLP support and localized global deployment for diverse regions.

GOAL: SCALING Z.AI TO SUPPORT 50+ FLOOD-PRONE URBAN CENTERS BY 2028

Summary


Z.AI EMERGENCY PORTAL: INTELLIGENT DISPATCH & RESCUE
SECTION 1 System Overview.
Victim Portal
Rescue Dashboard
Firebase
Real-time Sync
Connecting Victims & Rescuers via Mesh Simulation.
SECTION 2 Key Technical Features.
P0
P1
P2
AI
Agentic Priority Engine
AI analyzes SOS messages to auto-assign priority.
Hallucination Handling
Pre-validated inputs and checks prevent fake data.
NFR Stability
0% Error Rate & <800ms API Response.
SECTION 3 How It Works.
SUBMIT SOS
Describe emergency & location.
AI ANALYSIS
Priority & Location parsing.
DISPATCH
View on Rescue Dashboard.
SECTION 4 Risk Assessment & Mitigation.
Likelihood
Severity
API Timeout (High Score: 15)
Mitigation: Local Data Buffering
High Concurrency (Critical Score: 16)
Mitigation: Session State & Caching
SECTION 5 Quality Thresholds (CI/CD Gates)
Integration (Merge)
Automatic Build
100% Unit Tests
0 Linting Errors
85% Test Coverage
Deployment (Production)
90% Regression Rate
80% AI Pass Rate
Zero P0/P1 Bugs
<800ms Performance

Thank You For Listening

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Victim Portal: zai-flood-app.streamlit.app

Rescue Dashboard: zai-rescue-app.streamlit.app